I_Phi Landscape Analysis — L=4 Ne=4 U=4.0

```
System
======================================

  L=4  Ne=4
  t=1.000  U=4.0
  mode=uniform
  n = N = 4  (pure state)

  E0 = -2.102748483462077
  E1 = -1.8064238518231022

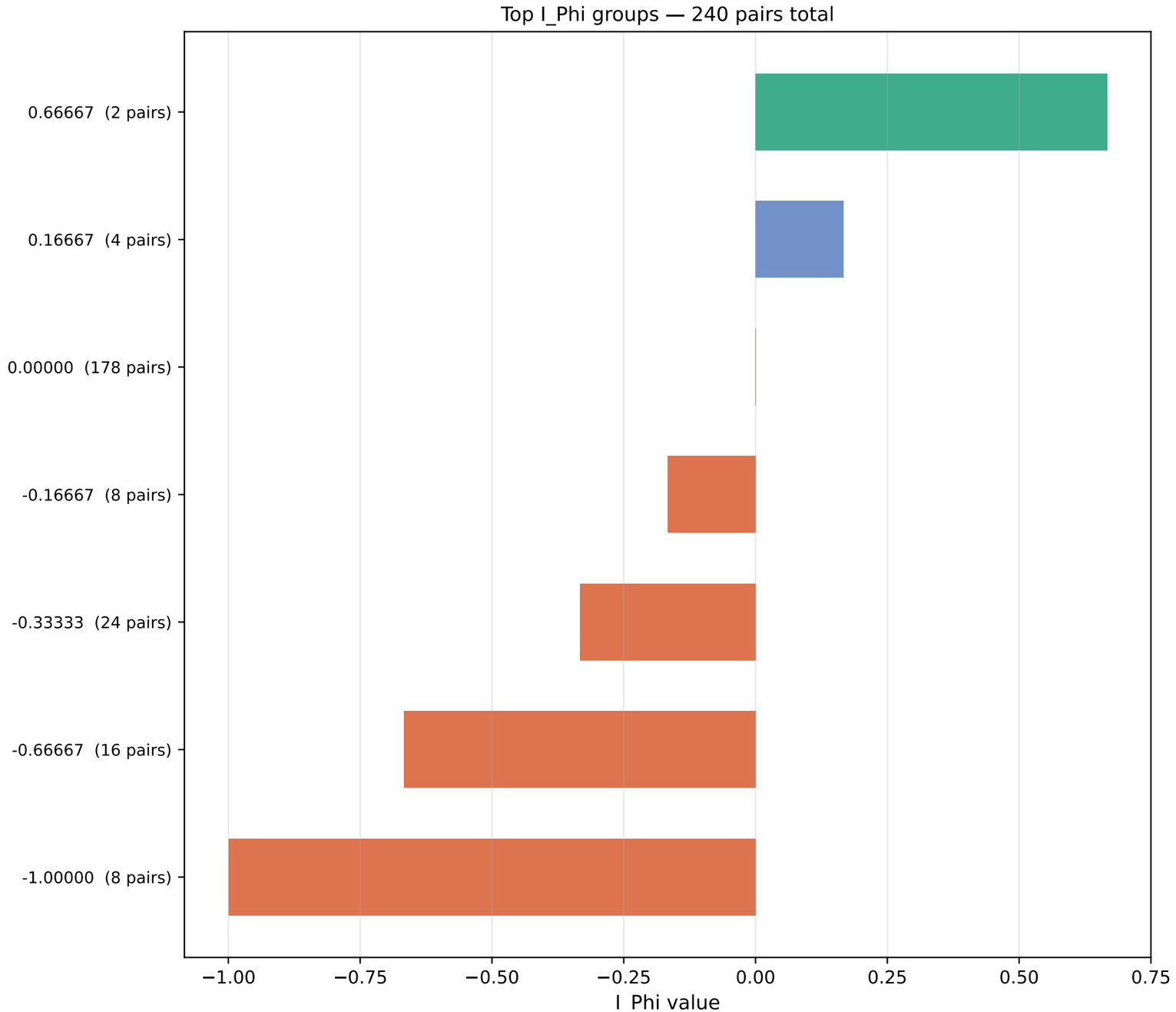
GME Results
======================================

  C_GME (exact eq.1)   = 0.91287093
  C_LB  (basis states) = 0.66666667
  Gap  = 0.24620426

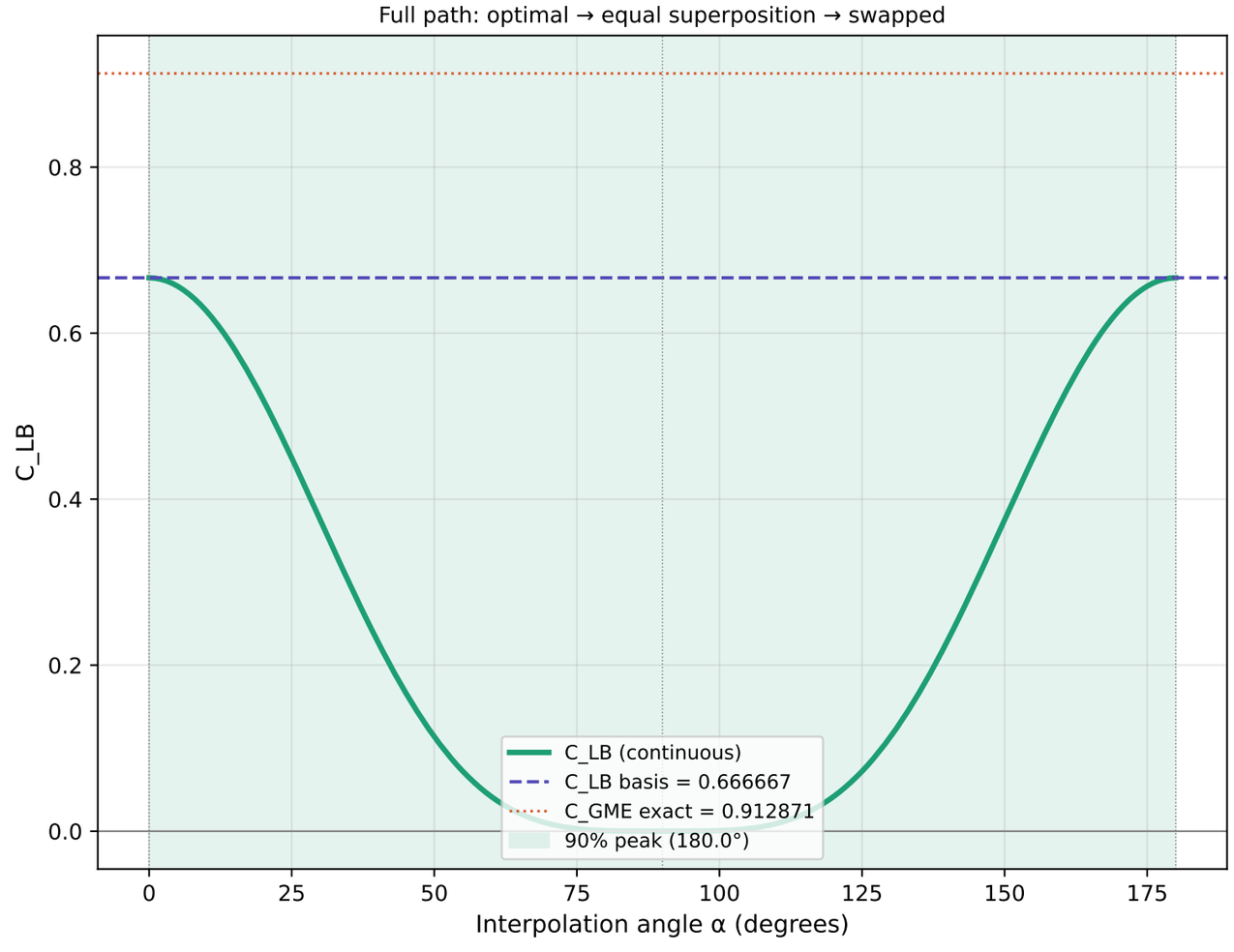
Optimal |Phi>
======================================

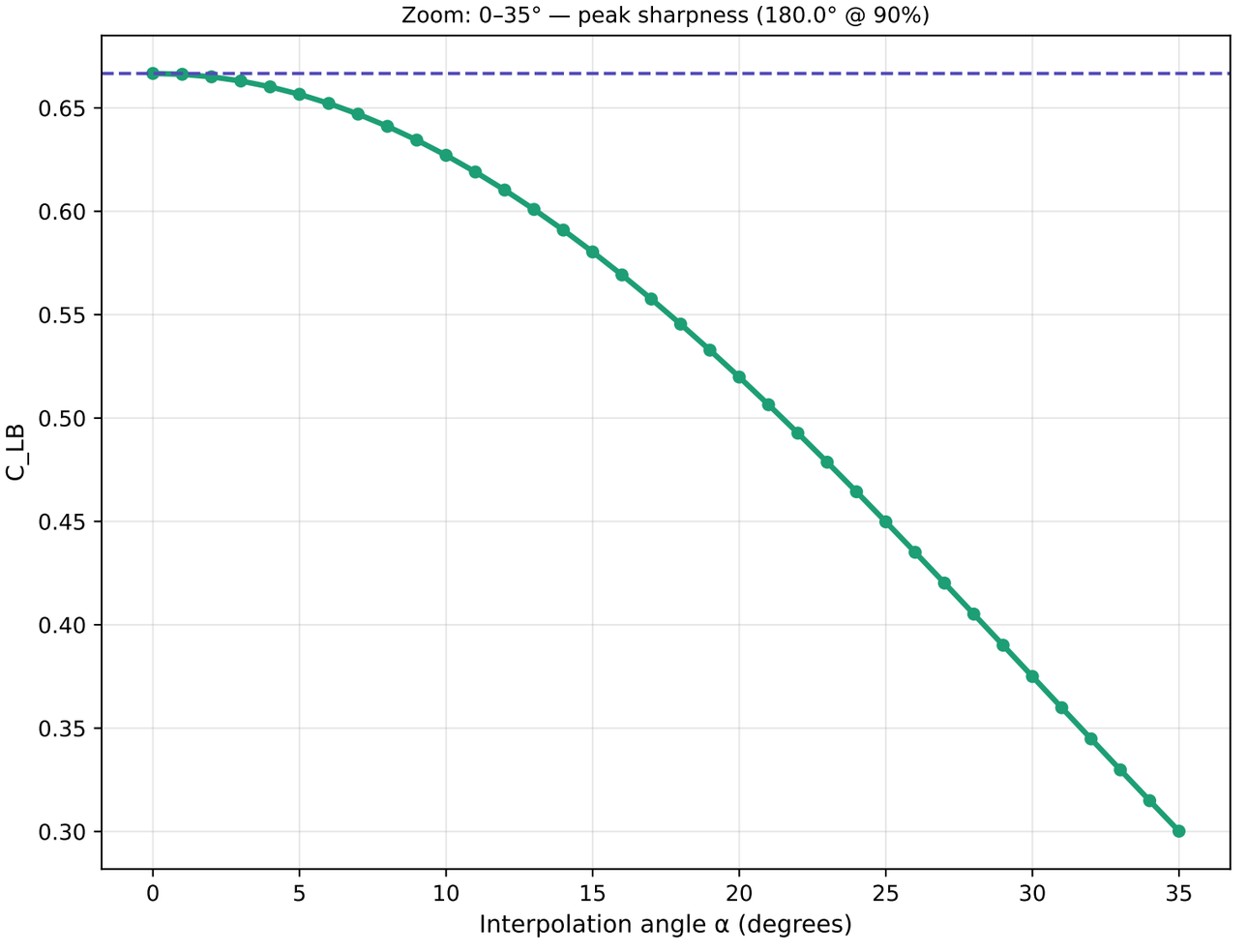
  z1 = |udud⟩
  z2 = |dudu⟩

  Pairs at max     : 2
  Physically unique: 1
```



Continuous Landscape: I_Phi along Bloch-sphere interpolation path





Landscape Summary: 2D Surface + Supervisor Conclusions

Supervisor's Question

Are the maxima only at basis states, or elsewhere?

DISCRETE SEARCH

Pairs at max: 2
Unique: 1
-> SINGLE optimal point

CONTINUOUS LANDSCAPE

Peak width @90%: 180.0°
-> BROAD

CONCLUSION

Basis state is the UNIQUE global optimum. No other family of states achieves the same I_Phi value.

